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原　著

# リゾチーム - キトサン糖複合体（LYZOX）とデカン酸含有スプレーをマラセチア皮膚炎が疑われる犬の皮膚病変部に用いた効果の検討：9 症例

# Evaluation of the effect of a lysozyme-chitosan sugar complex (LYZOX) and decanoic acid spray on local lesions of nine dogs suspected of Malassezia dermatitis

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**Abstract:** *Malassezia* Dermatitis is a common skin disease in dogs and likely to recur. Conventionally, azole-based antifungals have been used systemically or topically, however their use has been associated with problems of drug resistance and hepatotoxicity, and the shampoo therapy can be very difficult for some patients and owners (at home). There is a skin conditioning spray product which is used mainly by human care facilities which contains D-LYZOX as the chief active ingredient. We evaluated its clinical effectiveness against local lesions on nine dogs with suspected *Malassezia* Dermatitis, applying it twice a day for three weeks. One case had mild erythema at the start of treatment, but natural recovery occurred immediately. The dermatitis severity score and PVAS of the spray-treated dogs significantly decreased ($P<0.01$). The counts of *Malassezia* spp. in the lesions were lower, but the results were not statistically significant. In order to verify the contribution of the anti-inflammatory property of this formulation to the improvement of the dermatitis severity score and PVAS of the cases, we examined the effects of D-LYZOX on neutrophil adhesion as an *in vitro* parameter of inflammatory reaction. D-LYZOX inhibited lipopolysaccharide-induced adhesion of neutrophils obtained from a normal dog, suggesting the anti-inflammatory activity of D-LYZOX. These results suggest that this spray formulation containing D-LYZOX can improve the PVAS and dermatitis severity score of suspected local *Malassezia* dermatitis in dogs. We also discussed the need to clarify these clinical effects and their association with the suppression of neutrophil function.

**Key words:** dog, D-LYZOX, *Malassezia*, local, treatment

**要約**：マラセチア皮膚炎は犬では一般的な皮膚疾患であるが，再発性となりやすい。従来のアゾール系抗真菌薬の内服・外用療法は，耐性菌出現や肝毒性が問題であり，シャンプー療法は簡便とは言えない。そこで今回ヒトの介護現場で用いられているスプレー剤に着目し，犬 9 頭のマラセチア皮膚炎が疑われる皮膚病変部に 1 日 2 回ずつ 3 週間用いて臨床的な効果を検討した。試験期間を通した有害事象として 1 頭に軽度の一過性の発赤が認められたが，すぐに自然に消退した。スプレー

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